# Daughter-Universe Nucleation from a Metastable Dark-Energy Vacuum: Thermal and Quantum Decay, Past-Light-Cone Survival, and Reheating Bounds

Leon Sandler
Independent Researcher, The Conscious Machine — Independent Research, Northbrook, Illinois, USA
sandler.leon@gmail.com · ORCID 0009-0007-4584-808X

**Abstract**

A metastable dark-energy vacuum could decay by nucleating bubbles that evolve into daughter regions. Whether such a region can be a hot expanding universe, and whether cosmological cooling changes the probability of the transition, are quantitative questions, and this paper answers them for an explicit model. For a scalar potential with a metastable minimum at the dark-energy scale, the O(4) Coleman–De Luccia bounce is computed in flat space and with gravity (shooting, an independent boundary-value solution, a numerical error budget, the negative-mode structure and 21 executable checks). All scales are set by the dark energy, so $g = f^2/M_{\rm Pl}^2 \sim 10^{-60}$, the Hawking–Moss channel is irrelevant, and the quantum decay is controlled by one action $B = b(\eta)/\lambda$, with $\Gamma/H^4 = 1$ at $B \simeq 287$. Survival of the present vacuum is evaluated on the Lorentzian past light cone of the observed expansion history, $V_4 = 0.059\, H_\Lambda^{-4}$, 445 times below the Euclidean four-sphere, which gives the weak 95% bound $\Gamma/H_\Lambda^4 < 51$. Cooling is added through a thermal mass: the thermally corrected potential is an exact rescaling of the zero-temperature family, with $\eta_{\rm eff} = \eta/(1-2\theta)^{3/2}$, and the O(3) thermal saddle dominates above a crossover temperature of 0.18 meV, below the CMB temperature today (0.23 meV) for the benchmark. The rate falls monotonically as the universe cools, so cooling cannot raise the late-time transition probability: a scalar decoupled from the CMB is unaffected, while one in equilibrium with it raises the excluded fraction of the prior from 15% to 24–49% and closes the late-time decay window. The fractions refer to stated priors and are not probabilities for our universe. The daughter region reheats to at most 2.2 meV, $4.6 \times 10^8$ times below nucleosynthesis; it stays curvature dominated, a hot epoch at 1 s would need a parent energy scale at least 41 orders of magnitude above the Planck scale, and a second field supplies no energy because the released energy is the potential difference. A matter trigger needs $\Delta \ln m \gtrsim 8.9 \times 10^{-4}$. The daughter region is cold and empty, and a cyclic reset by the present dark energy is not supported; consistency conditions and observational tests are listed separately.

**Keywords:** *false vacuum decay; Coleman–De Luccia instanton; thermal activation; daughter universe; dark energy; reheating; cyclic cosmology; past light cone; falsifiability*

# 1 Introduction

In the standard cosmological model the late universe approaches de Sitter space: matter and radiation dilute, and a positive vacuum energy $\rho_\Lambda$ remains. If that energy is not an exact cosmological constant but the energy of a metastable scalar vacuum, the vacuum can decay by quantum tunneling through a Coleman–De Luccia (CDL) instanton [1]. It has been suggested that such a late-time transition could reset the cosmological history, so that the long-term evolution is cyclic rather than terminal; cyclic and bouncing proposals are discussed in [2, 3, 4, 5]. Two things must be kept apart. The formation of a bubble universe is not a cyclic cosmology: a cycle needs a mechanism that connects successive epochs. The question tested here is narrower and can be answered by calculation: can the tunneling of the present dark energy produce a new hot, expanding epoch? That requires a specified metastable minimum, the tunneling action including gravity, a rate compared with the observed expansion history, and the interior of the nucleated region followed to see whether it can become a hot big bang.

A further distinction concerns the outcome. In a branching or recycling picture (Option A) a parent metastable vacuum nucleates daughter regions by tunneling; each is a bubble inside the parent spacetime, and whether it is a hot expanding universe, or a universe at all in the sense of [6, 7, 8], is a separate question. In a genuinely cyclic cosmology (Option B) the same region expands, contracts, bounces and reheats, which needs a dynamical bounce beyond tunneling and raises problems of stability and entropy [9, 10]. This paper addresses Option A with the Coleman–De Luccia theory and treats Option B only by stating, in Section 5.4, what it would require.

Four claims are tested. (i) A scalar potential with a metastable minimum at the dark-energy scale can nucleate bubbles of lower-energy vacuum at a rate that is neither negligible nor already excluded. (ii) The nucleated region reheats into a hot expanding universe. (iii) The transition is triggered, or enhanced, by the dilution of matter in the late universe. (iv) Cosmological cooling, acting on a scalar coupled to the CMB, makes the transition more probable at late times. The analysis is conditional: it takes the dark-energy density as the false-vacuum energy and asks what follows. The result is a quantitative consistency and exclusion analysis, not a new cosmological mechanism.

The contributions are the following.

- A reproducible map from the parameters of an explicit scalar potential to cosmological decay times, with a numerical error budget, an independent boundary-value solution, the negative-mode structure of the bounce, the thin-wall, Parke and Hawking–Moss limits, and a treatment of the prefactor as a systematic uncertainty (Sections 2–4).
- A joint analysis of nucleation, survival and the post-transition energy budget, with the survival of the present vacuum evaluated on the Lorentzian past light cone of the observed expansion history (Section 4.2), and a statement of which energy sources the reheating bound excludes by geometry and which only by energy conservation (Section 5).
- Explicit conditions that separate a late-time vacuum transition from a viable cosmological

restart, divided into internal consistency conditions and observational tests (Section 7), and a model-independent requirement on any matter coupling that could trigger the transition (Section 6).

- A finite-temperature extension: the exact rescaling of the thermally corrected potential, the O(3) and O(4) exponents and their crossover, the past-light-cone survival integral with a time-dependent rate, and prior maps showing that cooling cannot raise the late-time transition probability (Sections 4.6–4.8).

### 1.1 Relation to previous work

The ingredients are established, and the paper claims novelty only for the specific combination and the quantitative results for the dark-energy vacuum. Table 1 states for each ingredient what is known, what is used from the literature and what is computed here.

## 2 Scalar effective theory and the potential

We consider a real scalar field minimally coupled to Einstein gravity,

$$S = \int d^4x\sqrt{-g}\left[\frac{M_{\rm Pl}^2}{2}R - \frac{1}{2}g^{\mu\nu}\partial_\mu\varphi\,\partial_\nu\varphi - V(\varphi)\right] \tag{1}$$

where $M_{\rm Pl}$ is the reduced Planck mass. The false vacuum has the energy density $V_f = V(\varphi_f)$, which is identified with the observed dark energy, $V_f = \rho_\Lambda = 2.55\times10^{-47}\ \mathrm{GeV}^4$ (quarter-root 2.25 meV; Planck 2018 [37], $H_0 = 67.66\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_\Lambda = 0.6889$). The asymptotic de Sitter rate is $H_\Lambda = \sqrt{V_f/3}/M_{\rm Pl} = 1.20\times10^{-42}$ GeV, so that $H_\Lambda^{-1} = 17.4$ Gyr, and $M_{\rm Pl}^4/V_f = 1.38\times10^{120}$, with $\ln(M_{\rm Pl}^4/V_f) = 276.6$. A microscopic origin of the potential is not assumed.

A single dimensionless family covers the generic metastable case. With $x = \varphi/f$,

$$V(\varphi) = V_t + f^4\lambda\left[\frac{(x^2-1)^2}{8} + \frac{\eta\,x}{2} - p_{\rm min}(\eta)\right], \qquad x^3 - x + \eta = 0 \;\Rightarrow\; x_t < x_{\rm top} < x_f \tag{2}$$

where $p_{\rm min}$ is the value of the bracket at the true minimum, so that $V(\varphi_t) = V_t$. The tilt $\eta$ lifts the degeneracy of the two wells; a false minimum exists only for $\eta < \eta_c = 2/(3\sqrt{3}) = 0.3849$. The energy difference and the barrier height above the false minimum are

$$\Delta V = V_f - V_t = f^4\lambda\,\delta(\eta), \qquad V_{\rm top} - V_f = f^4\lambda\,\beta(\eta), \qquad \delta \simeq \eta\ (\eta \ll 1) \tag{3}$$

The motivation for this family is minimality. A quartic with a small linear term is the simplest renormalizable potential with a metastable minimum and was the standard example in the original treatments of vacuum decay [11, 1]; it has three parameters with direct meaning, the quartic coupling $\lambda$, the tilt-to-barrier ratio $\eta$ and the scale $f$, and the latter is fixed by the dark energy. The coupling is restricted to $10^{-2} \le \lambda \le 4\pi$, the upper limit being the perturbative-unitarity bound for a quartic interaction and the lower limit an arbitrary but wide cut that

Table 1: Closest literature and what this paper adds.

| Ingredient | Established (references) | This paper |
|---|---|---|
| Bounce, thin wall, gravity, Hawking–Moss | Coleman; Callan–Coleman; CDL; Hawking–Moss; Parke; Jensen–Steinhardt; Hackworth–Weinberg [11, 12, 1, 13, 14, 15, 16] | Used as validation limits. New: an explicit dark-energy-scale potential, $g \sim 10^{-60}$, and a numerical error budget and negative-mode check (Section 3). |
| Prefactor of the rate | Determinant methods [17, 18] | Not recomputed; treated as a systematic uncertainty with a stated spread and shown to enter $B_c$ additively (Section 4.3). |
| Survival from the past light cone | Standard for the electroweak vacuum [19, 20]; percolation [21, 22] | Applied to the dark-energy vacuum with the observed expansion history: $V_4 = 0.059\, H_\Lambda^{-4}$, 445 times below the Euclidean four-sphere (Section 4.2). |
| Open interior and reheating | Open universes from bubbles need an inflationary stage [23, 24, 25] | Quantified for the dark-energy scale: curvature parameter, e-folds and the parent scale needed for a hot epoch (Section 5.3). |
| Finite-temperature decay | Thermal masses [26, 27]; thermal activation and the O(3) bounce [28, 29, 30] | Exact rescaling of the tilted quartic family under a thermal mass; O(3)/O(4) crossover for the dark-energy vacuum; survival with $\Gamma(T_\gamma(t))$ (Sections 4.6–4.8). |
| Daughter universes by tunneling | [6, 7, 8] | The nucleated region quantified as a cold, curvature-dominated daughter region, with the energy budget and reservoir bounds (Section 5). |
| Cyclic and bouncing models | [2, 3, 4, 5] | These supply the reset by other means; here one proposed mechanism, tunneling of the present dark energy, is tested. |
| Landscape and bubble collisions | [31, 32, 25, 33, 34] | Observable signals need $\Gamma/H^4 = O(1)$; the width of that window in $B$ and its prior mass are computed (Section 4.5). |
| Matter couplings and screening | [35, 36] | A coupling-independent requirement, $\Delta \ln m$, and its translation to linear and quadratic couplings (Section 6). |

is varied in Section 4.5; the tilt is restricted to $0.03 \leq \eta \leq 0.38$, which excludes the nearly degenerate case, where $B \propto \eta^{-3}$ is so large that the transition is irrelevant, and ends just below the point where the minimum disappears. Other potentials change $b(\eta)$ and the interior energy fraction but not the scales that decide the results (Section 8). A scalar with a mass of the order of meV is technically unnatural without a protecting symmetry; this is a known property of every quintessence-like model and is not addressed here.

The choice $V_t = 0$ (decay to a Minkowski vacuum, $r_L \equiv V_f/\Delta V = 1$) is the baseline, because it maximizes the energy released to the interior for a given $V_f$ when $V_t \geq 0$; $V_t < 0$ (anti-de Sitter) and $V_t > 0$ are considered in Sections 5.3 and 4.5. The scale $f$ is not free: it is fixed by $V_f = \rho_\Lambda$, $f^4 = \rho_\Lambda/(r_L \lambda \delta)$. For $\lambda = 1$ and $\eta = 0.3$ one finds $f = 3.05$ meV and a field excursion of 5.8 meV, that is, the dark-energy scale. The gravitational parameter is then fixed too,

$$g \equiv \frac{f^2}{M_{\rm Pl}^2} = \sqrt{\frac{V_f/M_{\rm Pl}^4}{r_L\,\lambda\,\delta(\eta)}} \tag{4}$$

which takes the values $6.4 \times 10^{-61}$ to $2.7 \times 10^{-60}$ over the benchmark parameters of Table 4. Gravity is therefore a $10^{-60}$ effect in every equation below that does not contain the large number $M_{\rm Pl}^4/V_f$.

# 3 The Euclidean bounce, with and without gravity

## 3.1 Equations and action

The O(4)-symmetric Euclidean metric $ds^2 = d\xi^2 + \rho(\xi)^2 d\Omega_3^2$ gives the CDL equations [1]

$$\varphi'' + 3\frac{\rho'}{\rho}\varphi' = \frac{dV}{d\varphi}, \qquad \rho'^2 = 1 + \frac{\rho^2}{3M_{\rm Pl}^2}\left(\frac{1}{2}\varphi'^2 - V\right) \tag{5}$$

with $\varphi'(0) = 0$ and regularity at the second pole of the compact de Sitter geometry. On shell, the action of the bounce relative to the false vacuum is

$$B = S_E[\text{bounce}] - S_E[\text{false}], \quad S_E = 4\pi^2 \int d\xi\, \left(\rho^3 V - 3M_{\rm Pl}^2 \rho\right), \quad S_E^{\rm dS} = -\frac{24\pi^2 M_{\rm Pl}^4}{V_f} \tag{6}$$

In the flat-space limit $B = 2\pi^2 \int dr\, r^3\, [\frac{1}{2}\varphi'^2 + V - V_f]$, and for a thin wall $B = 27\pi^2\sigma^4/(2\Delta V^3)$ with $R_c = 3\sigma/\Delta V$, $\sigma = \int \sqrt{2V}\, d\varphi$ [11]. With the field in units of $f$ and lengths in units of $(\sqrt{\lambda} f)^{-1}$ the flat-space action is $B = b(\eta)/\lambda$ exactly, and gravity enters only through $g$.

The Hawking–Moss action is $B_{\rm HM} = 24\pi^2 M_{\rm Pl}^4 (1/V_f - 1/V_{\rm top})$ [13]; an oscillating CDL bounce requires $|V''_{\rm top}| > 4H^2$ [15, 16], which fails for sufficiently flat potentials or strong gravity.

## 3.2 Numerical method and validation

The bounce is found by shooting on the central field value with overshoot/undershoot classification (adaptive eighth-order Runge–Kutta, relative tolerance $10^{-11}$) [38, 39, 40], with the

equations integrated through the wall and the action accumulated along the solution. For the gravitational problem the tail of the bounce outside the wall follows the false-vacuum de Sitter geometry $\rho = \sin(H\xi)/H$ toward the second pole, and the difference from the background action is taken analytically there, which avoids the cancellation of terms of order $1/g^2$. In the thin-wall regime shooting is exponentially sensitive to the initial condition, so an independent collocation boundary-value solver is used for $\eta < 0.1$, and the two agree to $2 \times 10^{-9}$ where both apply. The analytic thin-wall formula with gravity [14] serves as a further check of the gravitational suppression. Table 2 lists the checks of the bounce; the complete suite has 21 checks, all executed by a script distributed with the paper.

Table 2: Executable consistency checks of the numerical bounce.

| Check | Result |
|---|---|
| Thin-wall limit: $B/B_{\text{thin}}$ as $\eta \to 0$ | 0.9943 at $\eta = 0.03$, 0.9773 at 0.06 (monotone) |
| Shooting versus boundary-value solver | max relative difference $1.8 \times 10^{-9}$ (7 cases) |
| Scaling $B = b(\eta)/\lambda$ | max relative difference $9.8 \times 10^{-13}$ |
| de Sitter action $-24\pi^2 M^4/V_f$ | relative difference $4 \times 10^{-16}$ |
| Friedmann constraint along the bounce | max residual $5.3 \times 10^{-12}$ |
| Gravitational suppression against the analytic thin-wall (Parke) value | agree to 1.5% in $B/B_{\text{flat}}$ (4 points) |
| Hawking–Moss consistency | 84 bounces found in 98 grid points; none with $\lvert V''\rvert < 4H^2$; none above $B_{\text{HM}}$ |
| Interpolation of $b(\eta)$ off the table | max relative error $6.6 \times 10^{-5}$ |
| Derrick identity $2T + 4U = 0$ for the bounce | residual $1.4 \times 10^{-7}$ at the worst of four tilts |

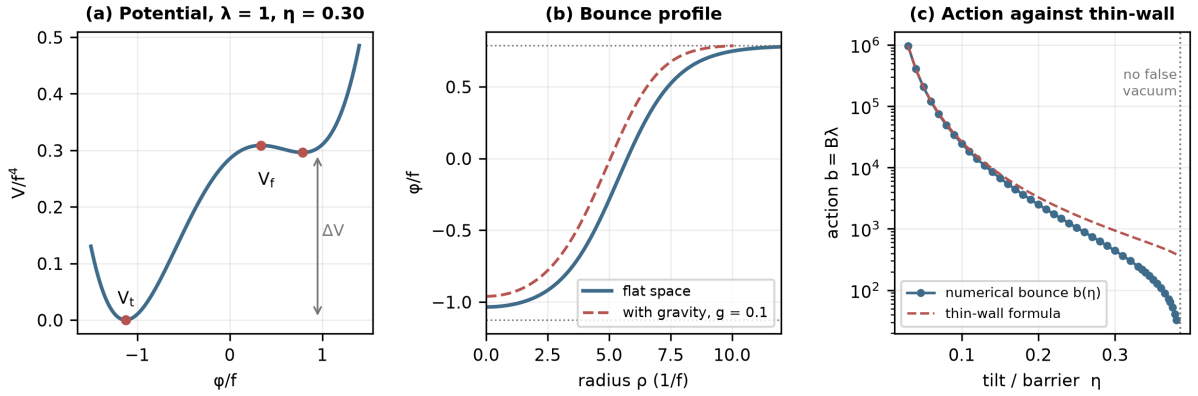



Figure 1: (a) The potential for $\lambda = 1$, $\eta = 0.30$. (b) The bounce profile in flat space and with gravity ($g = 0.1$). (c) The numerical action $b(\eta) = B\lambda$ against the thin-wall formula; the false minimum disappears at $\eta_c = 0.385$.

### 3.3 Error budget and negative modes

Table 3 lists how the flat-space action changes when each numerical choice is varied: the integrator tolerance and the integrator itself, the quadrature of the action integral, the tolerance of the bisection on the central field value, the truncation of the domain, and, for the boundary-value solution, the tolerance, the initial mesh and the padding of the domain. At the production settings the numerical uncertainty of $B$ is below $1 \times 10^{-8}$, many orders below the uncertainty of

the prefactor ($\delta \ln \Gamma = O(1)$) and below the precision of any quantity compared with the data. The only sensitive setting is the bisection: a tolerance of $10^{-6}$ in the central field value changes $B$ by up to $1 \times 10^{-4}$ in the thin-wall regime, which is why $10^{-13}$ is used.

Table 3: Numerical error budget of the flat-space action (largest relative change of $B$ over the tilts studied).

| Source | Range explored | Largest change |
|---|---|---|
| Integrator tolerance (DOP853) | $10^{-8}$ to $10^{-13}$ (production $10^{-11}$) | $3 \times 10^{-9}$ ($2 \times 10^{-11}$ for tolerance $\leq 10^{-10}$) |
| Integrator | DOP853, RK45, LSODA, Radau | $3 \times 10^{-10}$ |
| Quadrature of the action | $2 \times 10^4$ to $2 \times 10^6$ points | $2 \times 10^{-14}$ |
| Bisection on the central value | $10^{-6}$ to $10^{-13}$ (production $10^{-13}$) | $1 \times 10^{-4}$ ($1 \times 10^{-8}$ for tolerance $\leq 10^{-10}$) |
| Truncation of the domain | action cut at $r_{1/2} + k/m_f$, $k$ up to 16 | $1 \times 10^{-6}$ at the largest radius reached |
| Boundary-value solver: tolerance, mesh, padding | $10^{-6}$ to $10^{-10}$; 500 to 8000 nodes; 10 to 120 | $3 \times 10^{-12}$ |
| Shooting against boundary-value solution | $\eta = 0.10$ | $2 \times 10^{-9}$ |
| Gravity coefficient $\kappa(g \to 0)$ | integrators, tolerances, quadrature | $\kappa = 3.6407$, spread $3 \times 10^{-5}$ |

A CDL bounce is a saddle point with exactly one negative mode, which is what allows $\mathrm{Im}\, E$ to be interpreted as a decay rate [11, 12]. For $\eta = 0.05$, 0.15, 0.30 and 0.36 the spectrum of $-\nabla_4^2 + V''(\varphi_b)$ was computed in the angular sectors $l = 0, \ldots, 4$ (with $\psi = r^{-3/2}u$ the radial operator is $-u'' + [(l(l+2)+3/4)/r^2 + V'']u$, discretized on 30000 points). In every case the $l = 0$ sector has exactly one negative eigenvalue (lowest values $-0.002$, $-0.024$, $-0.21$, $-0.74$ in units of the false-vacuum mass squared), the $l = 1$ sector has the lowest eigenvalue consistent with the four translational zero modes (at most $2 \times 10^{-7}$ of $m_f^2$ in magnitude) and none is negative, no eigenvalue is negative for $l \geq 2$, the field is monotonic, so that $\varphi - \varphi_f$ has no node, and the Derrick identity holds to $1 \times 10^{-7}$. With gravity the negative-mode problem is subtle and has been discussed in the literature [41]; here $g \sim 10^{-60}$, the operator differs from its flat-space form by a relative $O(g)$, and the flat-space structure applies. The gravitational spectrum was not computed for large $g$.

Corrections of relative size $10^{-59}$ cannot be resolved in double precision, and the paper does not claim to resolve them. The flat-space action is computed with no gravity at all; the gravitational correction is the linear response $B = B_{\mathrm{flat}}(1 - \kappa g)$ with $\kappa(g \to 0) = 3.641$, obtained from $g = 10^{-3}$ and $2 \times 10^{-3}$ with the linear extrapolation $\kappa_0 = 2\kappa(10^{-3}) - \kappa(2 \times 10^{-3})$ (the two values differ by 2.6%, the curvature of $B(g)$) and stable to $3 \times 10^{-5}$ across integrators. At $g \sim 10^{-60}$ the correction $\kappa g \sim 9.7 \times 10^{-60}$ is some 59 orders of magnitude below every other uncertainty of the rate, so what matters is only that gravity enters as a small smooth perturbation, which the Hawking–Moss criterion confirms independently (Section 3.4).

### 3.4 Gravitational corrections and the Hawking–Moss limit

Fig. 2a shows the ratio $B(g)/B_{\rm flat}$ computed on a grid of $g$ and $\eta$. For small $g$ the suppression is linear, $B = B_{\rm flat}(1 - \kappa g)$ with $\kappa = 3.5$ to $6.7$ for $0.12 \le \eta \le 0.37$ (measured at $g = 10^{-3}$; below that the cancellation of the $1/g^2$ terms limits the numerical precision to about $10^{-4}$ in $B/B_{\rm flat}$). The suppression grows with $g$ as the bubble approaches the Hubble radius, and for $g \gtrsim 1$ no oscillating bounce is found: the numerical existence agrees with the condition $|V''_{\rm top}|/4H^2 > 1$ (Fig. 2b), which is satisfied everywhere a bounce was found.

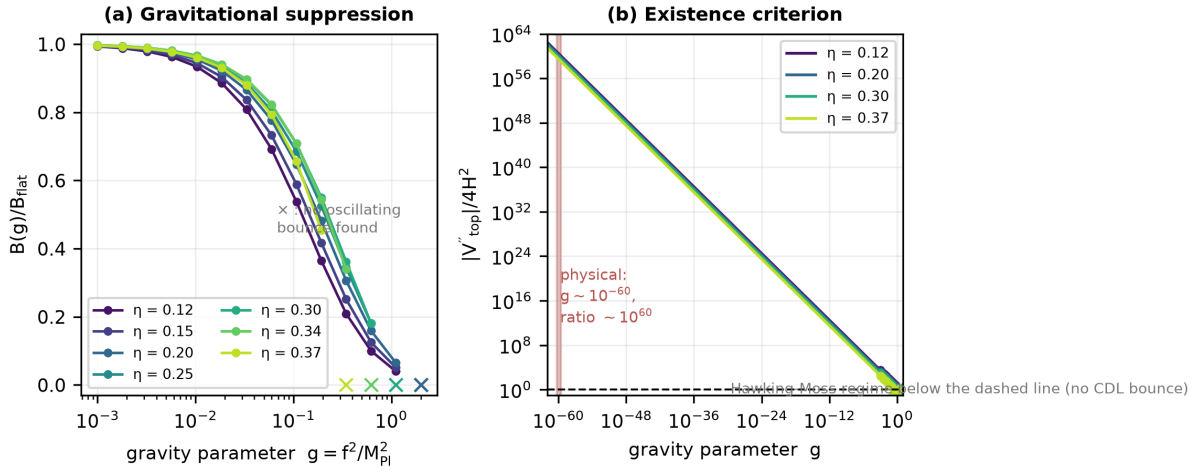



Figure 2: (a) Gravitational suppression of the action for several tilts; crosses mark the grid points at which no oscillating bounce was found. (b) The Hawking–Moss criterion $|V''_{\rm top}|/4H^2$ against $g$ (lines, analytic; markers, numerical bounce found/not found). The shaded band is the physical value fixed by the dark energy, $g \sim 10^{-60}$.

At the physical $g \sim 10^{-60}$ the relative gravitational correction is $\kappa g \sim 9.7 \times 10^{-60}$, the Hawking–Moss criterion is satisfied by a factor $\sim 10^{59}$–$10^{60}$, the Hawking–Moss action is $\ln B_{\rm HM} \simeq 279$ (an exponent of $10^{121}$), and the bubble radius is $\sim 2 \times 10^{-30}$ of the Hubble radius. Gravity and the Hawking–Moss channel play no role in a decay of the present dark energy within this class of potentials. They matter only if $f \gtrsim 0.3 M_{\rm Pl}$, which with $V_f$ fixed requires $V_f/f^4 \lesssim 9 \times 10^{-119}$, a barrier $\gtrsim 1 \times 10^{118}$ times the vacuum energy, and an action beyond any consequence for the transition.

## 4 Nucleation rate, cooling and survival in the observed universe

### 4.1 The rate

The rate per unit four-volume is $\Gamma/V_4 = A\, e^{-B}$ [11, 12]. In the rescaled variables $\Gamma/V_4 = K f^4 \lambda^2 e^{-B}$, where $K = (B/2\pi)^2 D$ contains the four translational zero modes and an unknown dimensionless factor $D$ from the remaining fluctuation determinant. Dividing by $H^4 = V_f^2/(9M_{\rm Pl}^4)$ and using $f^4 = V_f/(r_L \lambda \delta)$,

$$\frac{\Gamma}{H^4} = \frac{9K\lambda}{r_L\,\delta(\eta)}\,\frac{M_{\rm Pl}^4}{V_f}\,e^{-B(\lambda,\eta)} \tag{7}$$

The large number $M_{\rm Pl}^4/V_f = 1.38\times10^{120}$ is the number of Planck-volume cells in a Hubble four-volume. It compensates an action of $B \simeq \ln(M_{\rm Pl}^4/V_f) = 276.6$ plus the logarithm of the prefactor. Setting $\Gamma/H^4 = 1$ gives the critical action

$$B_c = \ln\frac{M_{\rm Pl}^4}{V_f} + \ln\frac{9K\lambda}{r_L\,\delta}, \qquad B_c \simeq 287 \;\; \text{(median over the prior; 5–95\% range 280–294)} \tag{8}$$

For a mean time to nucleation, an observer in the asymptotic de Sitter space sees the past-cone four-volume grow at the rate $(4\pi/3)H^{-3}$, so the mean time for one Hubble patch to nucleate a bubble is $t_{\rm dec} = H^{-1}/[(4\pi/3)\Gamma/H^4]$ for $\Gamma/H^4 \ll 1$. For $\lambda = 1$, $\eta = 0.30$ the action is $B = 441$ and the mean decay time is $10^{76}$ yr; at $\eta = 0.32$ ($B = 301$) it is $10^{15}$ yr; at $\eta = 0.34$ ($B = 194$) the rate is so large that the vacuum would not have survived and the model is excluded (Table 4).

Table 4: Benchmark potentials with the dark-energy density as false-vacuum energy ($r_L = 1$). $f$ is the field scale, $g = f^2/M_{\rm Pl}^2$, $\delta_B$ the relative gravitational correction to $B$, and $R_b/R_H$ the bubble radius over the Hubble radius.

| $\lambda$ | $\eta$ | $f$ (meV) | $B$ | $\ln\Gamma/H^4$ | $\log_{10} t_{\rm dec}$ (yr) | $g$ | $\delta_B$ | $\lvert V''_{\rm top}\rvert/4H^2$ | $R_b/R_H$ |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 0.30 | 3.05 | 441 | -153 | 76 | $1.6\times10^{-60}$ | $5.6\times10^{-60}$ | $5.3\times10^{59}$ | $2.6\times10^{-30}$ |
| 1 | 0.32 | 3.00 | 301 | -13 | 15 | $1.5\times10^{-60}$ | $5.4\times10^{-60}$ | $4.6\times10^{59}$ | $2.5\times10^{-30}$ |
| 1 | 0.34 | 2.96 | 194 | 93 | -31 | $1.5\times10^{-60}$ | $5.3\times10^{-60}$ | $3.8\times10^{59}$ | $2.3\times10^{-30}$ |
| 0.3 | 0.34 | 4.00 | 645 | -357 | 165 | $2.7\times10^{-60}$ | $9.7\times10^{-60}$ | $2.1\times10^{59}$ | $3.2\times10^{-30}$ |
| 3 | 0.25 | 2.42 | 349 | -60 | 36 | $9.9\times10^{-61}$ | $3.8\times10^{-60}$ | $1.2\times10^{60}$ | $2.3\times10^{-30}$ |
| 10 | 0.18 | 1.94 | 363 | -72 | 41 | $6.4\times10^{-61}$ | $3.2\times10^{-60}$ | $2.9\times10^{60}$ | $2.2\times10^{-30}$ |

### 4.2 Survival of the present vacuum: the past light cone

A bubble nucleated at an event $p$ expands at essentially the speed of light and reaches an observer at $O$ if and only if $p$ lies in the causal past of $O$. If nucleations are Poisson distributed with rate $\Gamma$ per unit four-volume, the probability that none has occurred in the past light cone of the present observer is [19, 20]

$$P_{\rm surv} = e^{-\Gamma V_4}, \qquad V_4 = \frac{4\pi}{3}\int_0^{t_0} dt\, a(t)^3\chi(t)^3, \qquad \chi(t) = \int_t^{t_0}\frac{dt'}{a(t')} \tag{9}$$

with $\chi$ the comoving distance of the light cone at time $t$ in a flat universe. A common order-of-magnitude shortcut uses the Euclidean volume of the de Sitter four-sphere, $8\pi^2/(3H^4)$; that is not the four-volume of any region of the observed spacetime, and it overstates $V_4$ by the factor given below. The observed history is $H^2 = H_0^2(\Omega_m a^{-3} + \Omega_r a^{-4} + \Omega_\Lambda)$ with $\Omega_\Lambda = 0.6889$, $\Omega_r = 9.14\times10^{-5}$ (from $T_{\rm CMB}$ and $N_{\rm eff} = 3.046$) and $\Omega_m = 0.3110$. Two independent quadratures give $V_4 = 0.12457\,H_0^{-4} = 0.0591\,H_\Lambda^{-4}$, agreeing to $1\times10^{-13}$; the result is converged in the lower cutoff of the scale factor and in the grid to $10^{-11}$ and reproduces the analytic Einstein–de Sitter

value to $2 \times 10^{-13}$ (check V14). The Euclidean volume is $26.3\, H_\Lambda^{-4}$, 445 times larger. Of $V_4$, 35% comes from events at redshift below 1, 73% below 2 and 97% below 5 (Fig. 3); the uncertainty from $\Omega_\Lambda$ at the Planck $1\sigma$ level is about 3.5%.

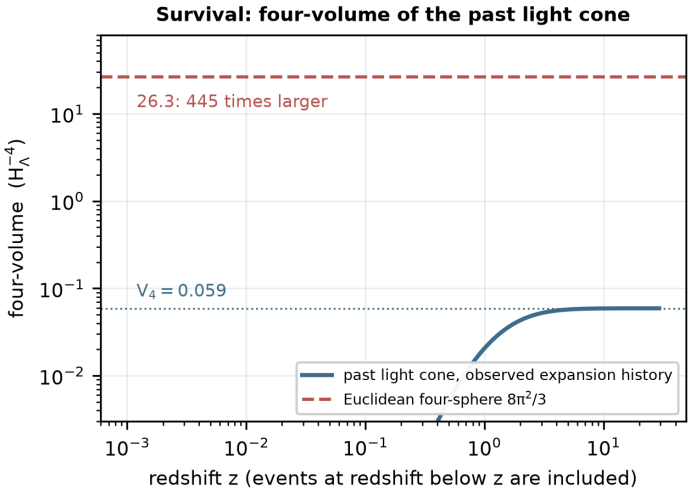



Figure 3: Four-volume of the past light cone of the present observer, accumulated over events at redshift below $z$, for the observed expansion history, against the Euclidean four-sphere volume.

The observation that no bubble has reached us is excluded at 95% confidence if $\Gamma V_4 > -\ln 0.05 = 3.00$, that is for $\Gamma/H^4 > 50.7$ or $B < B_c - 3.9$. This is a weak condition: it is 1334 times weaker than the Euclidean estimate ($\Gamma/H^4 > 0.038$), it moves the boundary of the allowed region by only $\Delta B = 7.2$ out of about 287, and it bounds the future mean decay time only by $t_{\rm dec} \gtrsim 8.2 \times 10^7$ yr (the formula for $t_{\rm dec}$ is used here as an order-of-magnitude statement, since $\Gamma/H^4$ is not small). Because the prior mass sits overwhelmingly at $B \ll B_c$ or $B \gg B_c$, the fraction of the prior excluded changes from 15.0% (Euclidean estimate) to 14.8% (Section 4.5). The criterion assumes that $\Gamma$ was constant, with the scalar in its false minimum throughout; it does not require observers to be present before the transition, and it says nothing about bubbles nucleated outside our past cone. Percolation of bubbles in de Sitter space requires a rate of order unity in Hubble units [21, 22], since otherwise the false-vacuum volume grows without bound and no wall collides with another. With the corrected survival bound this no longer conflicts with survival: the window $1 \lesssim \Gamma/H^4 < 51$ has the width $\Delta B = 3.9$ and contains 0.12% of the prior volume (0.18% for a percolation threshold of 0.1 and 0.05% for 10).

### 4.3 The prefactor and the validity of the semiclassical rate

The factor $D$ is the ratio of the fluctuation determinants with the zero modes removed; methods to compute it for specific potentials exist [17, 18] and are not repeated here. It is treated as a systematic uncertainty. Because $\ln D$ enters $B_c$ additively, a change of $D$ by $n$ decades moves $B_c$ by $2.3\, n$: three decades give $\pm 7$ out of 287, six decades give $\pm 14$ (5%), and the prior mass of every regime changes by less than $10^{-3}$ over spreads of 0, 3 and 6 decades (Table 5). The semiclassical expansion is controlled by $1/B$, which is $3.5 \times 10^{-3}$ at $B_c$, so that higher-loop corrections change $\ln \Gamma$ by $O(1)$ and not $O(B)$, which the prefactor spread already covers. The

regime in which the approximation fails is $B = O(1)$ to $O(10)$, where the wall is not thin on the scale of the field fluctuations; 0.5% of the prior has $B < 10$ and 2.5% has $B < 30$, and none of these draws survives the survival condition (all have $\Gamma/H^4 \gg 51$), so the failure of the semiclassical approximation affects only the excluded region and cannot change the classification of any draw that is allowed.

### 4.4 The falsification diagram

Fig. 4 maps the decay time over the $(\eta, \lambda)$ plane for $r_L = 1$. The region excluded by survival and the region of effective stability ($t_{\rm dec} > 10^{100}$ yr) are separated by a band: the decay time runs from the survival bound of $8.2 \times 10^7$ yr to $10^{20}$ yr within a change of $\Delta B \simeq 28$ in the action and to $10^{100}$ yr within $\Delta B \simeq 212$. The model is thus either excluded, or it describes a vacuum that decays on a time scale of $10^{20}$ yr or longer unless $B$ is tuned to within about 28 of the survival bound, that is within 10% of its value.

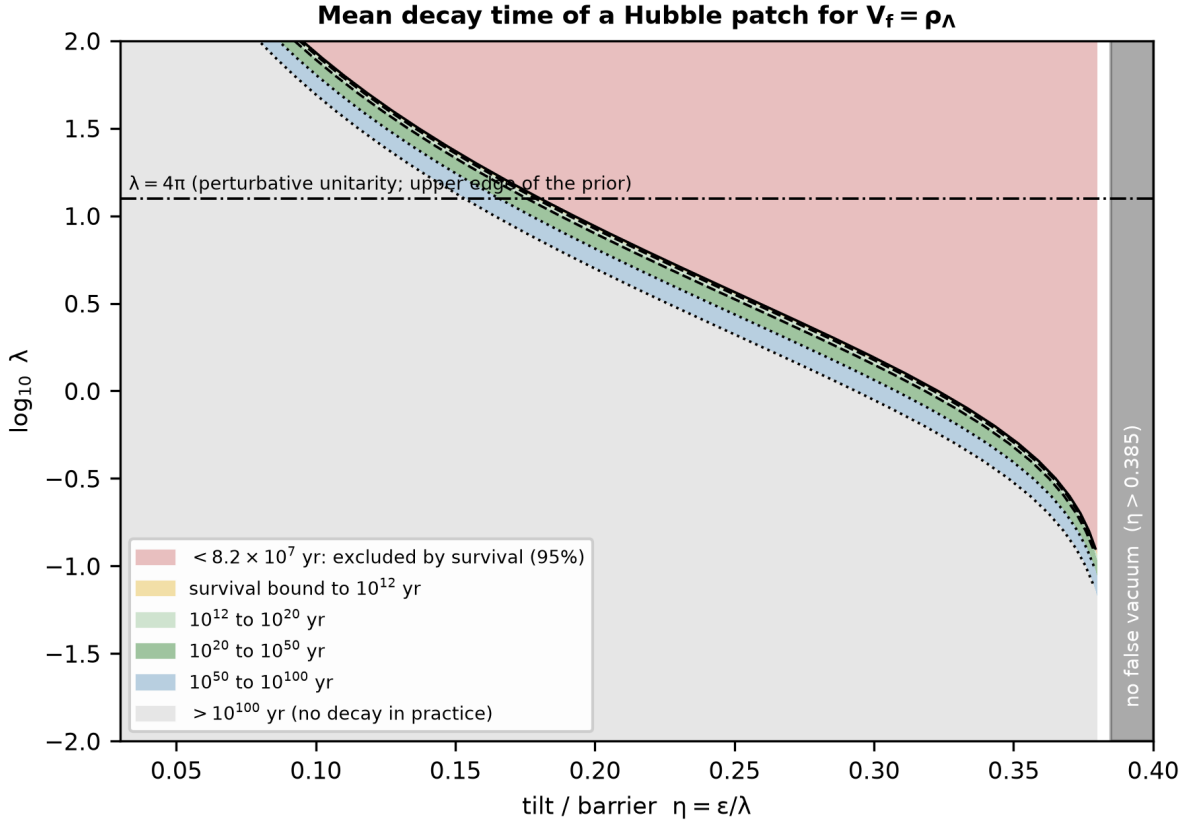



Figure 4: Mean decay time of a Hubble patch in the $(\eta, \lambda)$ plane for $V_f = \rho_\Lambda$ and a Minkowski true vacuum. Contours: survival bound ($8.2 \times 10^7$ yr), $10^{12}$, $10^{20}$, $10^{50}$ and $10^{100}$ yr. The upper edge $\lambda = 4\pi$ is the perturbative-unitarity limit used for the prior.

### 4.5 Prior-sensitivity Monte Carlo

To quantify how much of the parameter space the window occupies, 1,000,000 parameter sets are drawn from physically consistent priors ($\log_{10}\lambda$ uniform on $[-2, \log_{10} 4\pi]$, $\eta$ uniform on

$[0.03, 0.38]$, $\ln D$ Gaussian with a spread of 1.5 decades; the false minimum exists for every draw) and each is converted to a rate with the interpolated numerical action. The regime fractions are given in Table 5 together with batch-means standard errors (20 batches), convergence with the number of draws (Fig. 5c) and sensitivity to the priors. 14.8% of the prior volume lies above the survival bound, 0.77% decays within $10^{20}$ yr, 4.8% within $10^{100}$ yr and 80.4% shows no decay within $10^{100}$ yr. The fractions do not depend on the prefactor and the choice of $r_L$ is likewise immaterial. They do depend on the prior range of $\lambda$ and $\eta$ (Table 5; for $\lambda \leq 1$ the excluded fraction falls to 2.8%). They are therefore fractions of the stated prior, a property of the parameter space of this model and not measured probabilities, and they do not indicate how likely a transition is in our universe. $V_t < 0$ would enter through $r_L = 1/(1 + |V_t|/V_f) < 1$ and shifts $B_c$ upward by $\ln(1 + |V_t|/V_f)$; the case $r_L \geq 1$ is the one sampled.

Table 5: Prior mass (%) of each regime. Decay fractions are cumulative and refer to survival-consistent draws; the last column is the fraction with no decay within $10^{100}$ yr. These are fractions of the stated prior, not probabilities for our universe.

| Prior | above survival bound | decays by $10^{12}$ yr | by $10^{20}$ yr | by $10^{100}$ yr | no decay in $10^{100}$ yr |
|---|---|---|---|---|---|
| baseline | $14.77 \pm 0.04$ | $0.269 \pm 0.005$ | $0.77 \pm 0.01$ | $4.78 \pm 0.02$ | $80.4 \pm 0.0$ |
| $\sigma_{\log D} = 0$ | 14.74 | 0.270 | 0.76 | 4.80 | 80.5 |
| $\sigma_{\log D} = 3$ | 14.75 | 0.257 | 0.75 | 4.79 | 80.5 |
| $\sigma_{\log D} = 6$ | 14.74 | 0.267 | 0.76 | 4.79 | 80.5 |
| $r_L$ logU(1, 10) | 14.71 | 0.264 | 0.76 | 4.80 | 80.5 |
| $\lambda \leq 1$ | 2.76 | 0.122 | 0.35 | 2.50 | 94.7 |
| $\eta$ in [0.10, 0.38] | 18.45 | 0.323 | 0.95 | 5.96 | 75.6 |
| log10 $\lambda$ in [-4, 2] | 18.18 | 0.190 | 0.55 | 3.35 | 78.5 |

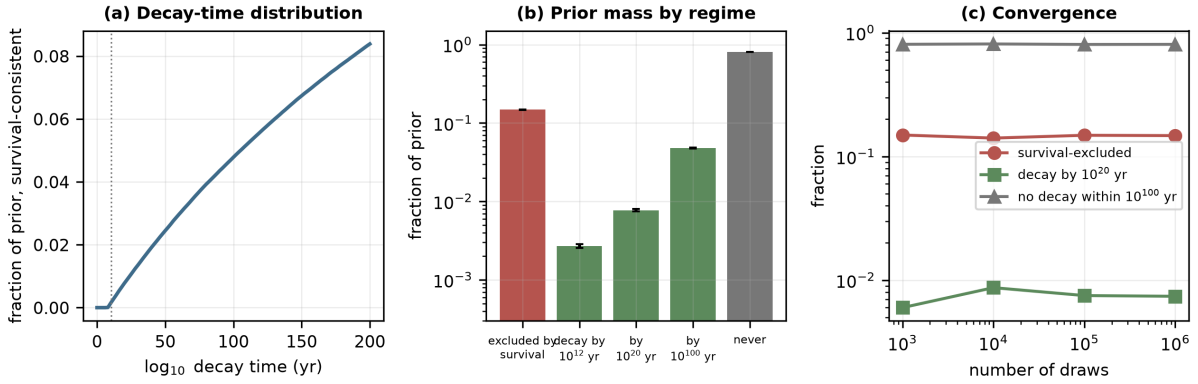



Figure 5: (a) Cumulative distribution of the decay time over survival-consistent draws. (b) Prior mass by regime with $3\sigma$ batch errors. (c) Convergence of the regime fractions with the number of draws.

### 4.6 Cooling and the thermally corrected potential

The calculation above treats the universe as a clock: the CMB temperature $T_\gamma = T_0/a$ falls with the expansion and enters nothing. It can enter the decay only if the scalar interacts with a thermal bath, and then through a thermal correction to the potential whose coefficient $c$ is fixed by those interactions and is not a universal constant. Two regimes are therefore kept

apart throughout: the scalar is decoupled and the decay is the zero-temperature O(4) channel of the previous sections, or it is in thermal equilibrium with the CMB, in which case the leading high-temperature term is

$$V_{\rm eff}(\varphi,T) = V_0(\varphi) + \tfrac{1}{2}\,cT^2\varphi^2, \qquad \theta \equiv \frac{cT^2}{\lambda f^2} \tag{10}$$

The thermal term is a rescaling of the zero-temperature family, which makes the analysis exact. With $x = sy$ and $s = \sqrt{1-2\theta}$ the potential $(x^2-1)^2/8+\eta x/2+\theta x^2/2$ becomes $s^4[(y^2-1)^2/8+(\eta/s^3)y/2]$ up to a constant. Hence a false minimum exists only for

$$\eta_{\rm eff} \equiv \frac{\eta}{(1-2\theta)^{3/2}} < \eta_c \quad\Longleftrightarrow\quad \theta < \theta_c(\eta) = \frac{1}{2}\left[1-\left(\frac{\eta}{\eta_c}\right)^{2/3}\right] \tag{11}$$

and the bounce actions are those of the zero-temperature table at the shifted tilt: $B_4 = b(\eta_{\rm eff})/\lambda$ and $S_3 = (fs/\sqrt{\lambda})\,s_3(\eta_{\rm eff})$, where $s_3 = 4\pi\int\rho^2[x'^2/2 + w - w_f]d\rho$ is the O(3) action at $\lambda = 1$, $\theta = 0$ (tabulated here for the first time for this family, with the boundary-value solver for $\eta < 0.1$, and tending to the thin-wall value $16\pi\sigma^3/3\Delta V^2$ as $\eta \to 0$: the ratio is 0.995 at $\eta = 0.03$). Direct solutions of the O(4) and O(3) bounce with the thermal term in the potential agree with these scalings to $2\times10^{-4}$ (check V19). For the benchmark $\lambda = 1$, $\eta = 0.30$: $f = 3.05$ meV, $\theta_c = 0.0765$, and today ($T_0 = 0.235$ meV) $\theta = 0.0059\,c$.

Cooling relaxes the thermal term, and the thermal history has a consequence that does not depend on the decay rate: the false minimum disappears at $T_c = f\sqrt{\lambda\theta_c/c}$, which is 2.7 meV (redshift 10) for $c = 0.1$ and 0.84 meV (redshift 2.6) for $c = 1$. A scalar in equilibrium with the CMB at higher temperatures would have settled into the true vacuum at the symmetric-to-broken transition, and the present dark energy would not be the false-vacuum energy. The observed universe in the false vacuum therefore requires that the scalar either decoupled before $T_c$ or has $c < 1.1\times10^{-5}$ if the bath is coupled all the way back to recombination ($z = 1100$). The high-temperature form is an upper bound on the effect for particles heavier than $T$ in the loop, since the true correction is then Boltzmann suppressed; the results below are therefore the largest thermal effect that the coefficient $c$ can produce.

### 4.7 The cooling history and the three decay channels

The temperature follows the observed history, $T_\gamma(a) = T_0/a$ with $H(a)$ of Section 4.2, and cosmic time $t(a) - t_0 = \int_1^a da'/[a'H(a')]$. Three channels compete. The Hawking–Moss transition is the horizon-scale channel of de Sitter space; its action is $\ln B_{\rm HM} \simeq 279$ (Section 3.4) and it plays no role. The de Sitter horizon temperature $T_{\rm dS} = H_\Lambda/2\pi = 1.91\times10^{-31}$ meV is $8.1\times10^{-31}$ of $T_0$, would equal the CMB temperature only after 69 e-folds ($\sim 10^{12}$ yr), and is not a homogeneous radiation bath; it is therefore kept separate and not combined with $T_\gamma$ by a maximum. The thermal exponent at the horizon temperature, $S_3/T_{\rm dS} = 4.2\times10^{32}$, rules out any activation by it. That leaves quantum tunneling (the O(4) bounce with action $B_4$) and thermal activation (the O(3) bounce with $S_3/T$). At a given temperature the saddle with the smaller exponent dominates, and the rate takes its exponent and prefactor; the two rates are not added,

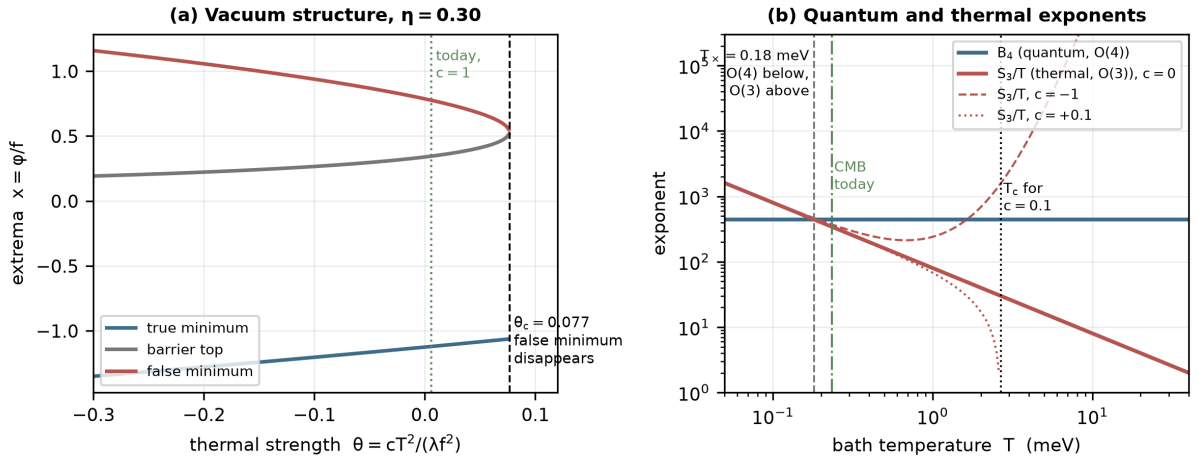



Figure 6: (a) Extrema of the thermally corrected potential for $\eta = 0.30$ against the thermal strength $\theta$; the false minimum disappears at $\theta_c$. (b) The quantum exponent $B_4$ and the thermal exponent $S_3/T$ against the bath temperature for the benchmark; the O(3) saddle dominates above the crossover temperature $T_\times$ (where $S_3/T = B_4$) and the O(4) saddle below. Solid: $c = 0$ (equilibrium without thermal mass); dashed: $c = -1$; dotted: $c = +0.1$.

because they are the two ends of one family of periodic instantons. The thermal prefactor is $\Gamma_T = D_T\, T^4 (S_3/2\pi T)^{3/2} e^{-S_3/T}$ with $D_T$ of order unity (a systematic of 1.5 decades, as for the quantum prefactor).

For the benchmark the crossover is at $T_\times = 0.181$ meV, so that today the thermal saddle already dominates ($S_3/T_0 = 341 < B_4 = 441$), the crossover being reached when the CMB has cooled by a factor 1.29, and in the past the thermal exponent was far smaller (Table 6, Fig. 6b). Cooling lowers the rate for $c \geq 0$ at every epoch: the thermal exponent grows as $1/T$ and the thermal mass relaxes, so $\ln\Gamma$ falls monotonically with time (check V20). The late-time decay is therefore the zero-temperature O(4) decay of Section 4, because the thermal contribution switches off exponentially fast as $T \propto e^{-H_\Lambda t}$. For $c < 0$ (symmetry non-restoration, outside the usual sign of the thermal mass) the barrier grows with $T$ and cooling lowers $B_4$, so the sign of the effect is reversed for the quantum channel, but the thermal exponent still has a minimum near $z \simeq 2$ for $c = -1$ and the past rate is not small.

Table 6: Exponents and rates for the benchmark ($\lambda = 1$, $\eta = 0.30$) as the universe cools. $B_4$ and $S_3/T$ at $c = 0$; $\ln\Gamma/H_\Lambda^4$ for the decoupled scalar (O(4) only) and for the equilibrated scalar with the dominant saddle in brackets.

| $z$ | $T$ (meV) | $B_4$ | $S_3/T$ | decoupled | $c = 0$ | $c = 0.1$ | $c = -1$ |
|---|---|---|---|---|---|---|---|
| 0 | 0.235 | 441 | 341 | -153 | -65 (O(3)) | -62 (O(3)) | -95 (O(3)) |
| 0.5 | 0.352 | 441 | 227 | -153 | 50 (O(3)) | 54 (O(3)) | 4 (O(3)) |
| 1 | 0.470 | 441 | 170 | -153 | 107 (O(3)) | 113 (O(3)) | 46 (O(3)) |
| 2 | 0.705 | 441 | 114 | -153 | 165 (O(3)) | 173 (O(3)) | 66 (O(3)) |
| 3 | 0.939 | 441 | 85 | -153 | 194 (O(3)) | 205 (O(3)) | 49 (O(3)) |

### 4.8 Survival with a time-dependent rate and the thermal prior maps

With a rate that depends on the temperature, the expected number of nucleations in the past light cone is

$$I = \frac{4\pi}{3}\int_{t_i}^{t_0} dt\,\Gamma(T_\gamma(t))\,a(t)^3\chi(t)^3, \qquad P_{\rm surv} = e^{-I} \tag{12}$$

with $\chi$ as in Eq. (9). It assumes a flat universe and a homogeneous rate per physical four-volume, and walls at the speed of light. For a constant rate it reduces to Eq. (9), reproduced by the quadrature to $7\times10^{-9}$. For the benchmark, $\ln I = -155$ for the decoupled scalar (the vacuum survives), while an equilibrated scalar gives $\ln I = 99$ ($c=0$, bath coupled for $z\le1$), 186 ($z\le3$) and 240 ($z\le10$): the benchmark is excluded by many orders of magnitude as soon as it is in equilibrium with the CMB. The same calculation over the priors of Section 4.5 ($2\times10^5$ draws, the same seed, plus a thermal prefactor spread of 1.5 decades) gives Table 7 and Fig. 7.

Table 7: Prior mass (%) by thermal scenario ($2\times10^5$ draws; $z$ is the redshift below which the scalar is in equilibrium with the CMB). "Inconsistent" draws have no false minimum at some redshift of the equilibrium epoch. Decay fractions refer to the allowed draws and to the zero-temperature decay of the future.

| Scenario | above survival bound | no false minimum in the past | allowed | decays by $10^{20}$ yr | no decay ir |
|---|---|---|---|---|---|
| decoupled | 14.7 | 0.0 | 85.3 | 0.79 | 80.5 |
| $c=0$, $z\le1$ | 24.2 | 0.0 | 75.8 | 0.00 | 75.8 |
| $c=0$, $z\le3$ | 33.9 | 0.0 | 66.1 | 0.00 | 66.1 |
| $c=0$, $z\le10$ | 49.3 | 0.0 | 50.7 | 0.00 | 50.7 |
| $c=0.1$, $z\le1$ | 24.9 | 1.3 | 73.8 | 0.00 | 73.8 |
| $c=0.1$, $z\le3$ | 33.6 | 7.4 | 59.0 | 0.00 | 59.0 |
| $c=1$, $z\le1$ | 23.2 | 17.6 | 59.2 | 0.00 | 59.2 |
| $c=-0.1$, $z\le1100$ | 38.0 | 0.0 | 62.0 | 0.00 | 62.0 |
| $c=-1$, $z\le1100$ | 23.8 | 0.0 | 76.2 | 0.17 | 75.4 |

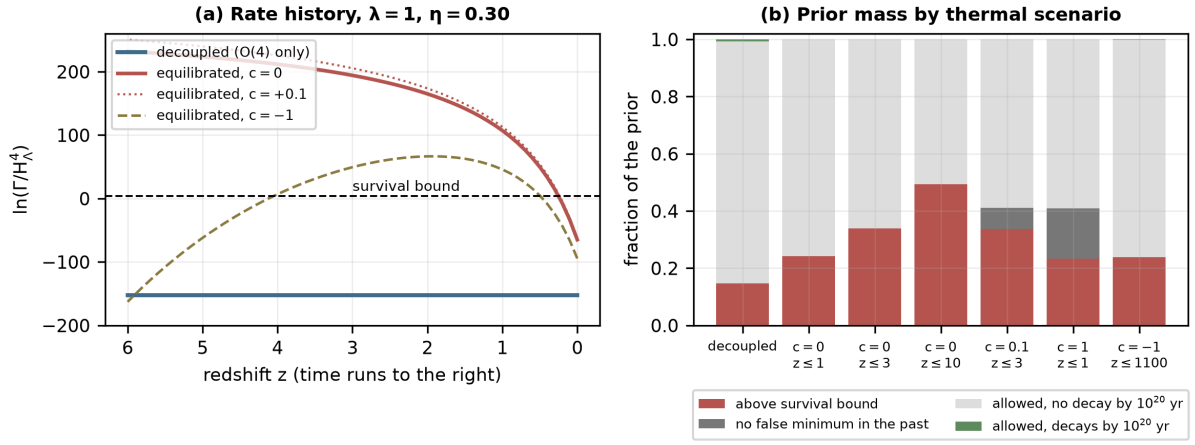



Figure 7: (a) Rate history of the benchmark: the decoupled O(4) rate is constant, the equilibrated rate falls as the universe cools; the dashed line is the survival bound on $\Gamma/H_\Lambda^4$. (b) Prior mass by thermal scenario.

The results answer the question posed by cooling. First, a scalar that is not coupled to the CMB

is unaffected; the decay of the previous sections stands. Second, a scalar in equilibrium with the CMB is constrained more strongly, not less: the prior mass above the survival bound rises from 14.7% to 24% ($z \leq 1$), 34% ($z \leq 3$) and 49% ($z \leq 10$) for $c = 0$, and a further 1.3% to 18% of the draws have no false minimum in the equilibrium epoch for $c = 0.1$ and 1. Third, in no equilibrated scenario with $c \geq 0$ does a survival-consistent draw decay within $10^{20}$ yr: the decay window of Section 4.4 closes completely. The reason is that the allowed draws must have a thermal exponent large enough at $z \sim 1$, and because $S_3/T < B_4$ above $T_\times$ this requires $B_4$ well above the critical action, so that the zero-temperature decay time of the future exceeds $10^{100}$ yr. Fourth, only the non-standard sign $c = -1$ leaves a window (0.17% of the prior against 0.79% for the decoupled scalar). In short, cooling cannot raise the probability of a late-time transition: it lowers the thermal channel monotonically, the late-time decay is the zero-temperature O(4) decay, and the thermal coupling only adds past constraints.

# 5 What the nucleated region can become

## 5.1 Energy budget

A thin-wall bubble of true vacuum is bounded by a wall that carries the released energy; its interior is, to leading order, the true vacuum with no excess energy to reheat. In the thick-wall regime relevant here the field at the centre of the bounce, $x_0$, lies above the true minimum, and the energy the interior can release is $\epsilon_{\rm in}\Delta V = V(x_0) - V_t$. Fig. 8c shows $\epsilon_{\rm in}$ against $\eta$: at $\eta = 0.30$, $\epsilon_{\rm in} = 0.018$, so 1.8% of $\Delta V$ resides in the interior and the remainder in the wall; $\epsilon_{\rm in}$ approaches unity only near $\eta_c$, where the barrier is gone.

## 5.2 Interior geometry and reheating

The interior of a CDL bubble has open spatial slices, $ds^2 = -dt^2 + a(t)^2 dH_3^2$ [23, 24], with the Friedmann constraint $(da/dt)^2 = 1 + a^2\rho/(3M_{\rm Pl}^2)$. Without a long inflationary stage inside the bubble (Section 5.3), the energy density is negligible against the curvature term at early times: $\alpha^2\rho \leq 1.1 \times 10^{-61}$ in units where $H_\Lambda = 1$, so the interior is a Milne universe, $a = t$. The condensate begins to oscillate at $t_0 = 1/m$ with $m = 3.61$ meV, behaves as matter and decays to radiation at a rate $\Gamma_\varphi \leq m$. With $\tau = H_\Lambda t$ the radiation density has the closed form $\rho_r(\tau) = \epsilon\,\tau_0^3\,\gamma\,\tau^{-4}\int_{\tau_0}^{\tau} s\,e^{-\gamma(s-\tau_0)}ds$ in units of $V_f$, which we compare with a numerical integration of the full system (relative difference $6\times10^{-7}$). The radiation temperature is $T = (30\rho_r/\pi^2 g_*)^{1/4}$ with $g_* = 3.36$.

Energy conservation bounds the reheating temperature independently of the details of the decay,

$$T_{\rm reh} \leq \left[\frac{30\,\epsilon_{\rm th}\,\epsilon_{\rm in}\Delta V}{\pi^2 g_*}\right]^{1/4}, \qquad \epsilon_{\rm th} \leq 1 \tag{13}$$

which for the largest release allowed when $V_t \geq 0$, $\epsilon_{\rm in}\Delta V = \Delta V \leq V_f = \rho_\Lambda$, gives $T_{\rm reh} \leq 2.19$ meV (25 K), against the CMB temperature of 0.235 meV today and $4.6\times10^8$ times below the $\sim 1$ MeV required for nucleosynthesis [42]. Equivalently, a thermal bath at 1 MeV needs an energy

density $1.4 \times 10^{35}$ times the dark-energy density. The closed form shows that the situation is worse than the bound: the radiation temperature after its maximum falls as $T \propto 1/\tau$ because the open interior keeps expanding, with $T\tau = 5.46 \times 10^{-43}$ GeV, so $T = 1$ MeV would have to occur at $t = 3.0 \times 10^{-22}$ s, before the condensate even starts to oscillate ($t_0 = 1.8 \times 10^{-13}$ s). Table 8 gives the maximum temperature for decay rates from $\Gamma_\varphi = m$ down to $10^{-20}m$ and the temperature one Hubble time after nucleation, $\sim 10^{-30}$ meV.

Table 8: Reheating of the bubble interior. $\epsilon$: fraction of $\Delta V$ in the interior; $\Gamma_\varphi/m$: decay rate of the oscillating condensate; $T_{\max}$: maximum radiation temperature; the fourth column is $T_{\max}$ over the bound of Eq. (13); the fifth is the temperature one Hubble time $1/H_\Lambda$ after nucleation; the last is $T_{\rm BBN}/T_{\max}$.

| $\epsilon$ | $\Gamma_\varphi/m$ | $T_{\max}$ (meV) | $T_{\max}/T_{\rm bound}$ | $T(1/H_\Lambda)$ (meV) | $T_{\rm BBN}/T_{\max}$ |
|---|---|---|---|---|---|
| 1 (upper bound) | 1 | 1.2445 | 0.57 | $8.7 \times 10^{-31}$ | $8.0 \times 10^{8}$ |
| 1 (upper bound) | 0.001 | 0.2318 | 0.11 | $4.1 \times 10^{-30}$ | $4.3 \times 10^{9}$ |
| 1 (upper bound) | 1e-09 | 0.0073 | 0.00 | $1.3 \times 10^{-28}$ | $1.4 \times 10^{11}$ |
| 1 (upper bound) | 1e-20 | 0.0000 | 0.00 | $7.3 \times 10^{-26}$ | $7.7 \times 10^{13}$ |
| 0.018 ($\eta = 0.30$) | 1 | 0.4558 | 0.57 | $3.2 \times 10^{-31}$ | $2.2 \times 10^{9}$ |
| 0.018 ($\eta = 0.30$) | 0.001 | 0.0849 | 0.11 | $1.5 \times 10^{-30}$ | $1.2 \times 10^{10}$ |
| 0.018 ($\eta = 0.30$) | 1e-09 | 0.0027 | 0.00 | $4.7 \times 10^{-29}$ | $3.7 \times 10^{11}$ |
| 0.018 ($\eta = 0.30$) | 1e-20 | 0.0000 | 0.00 | $2.7 \times 10^{-26}$ | $2.1 \times 10^{14}$ |

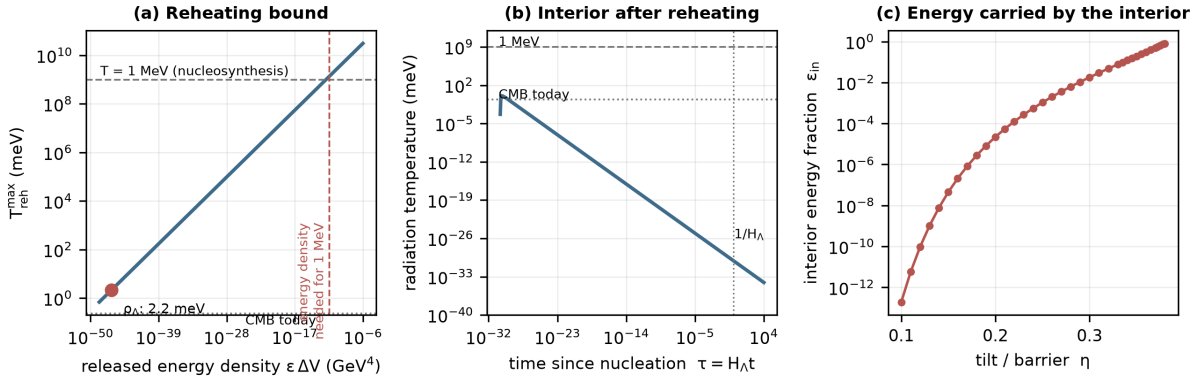



Figure 8: (a) Maximum reheating temperature against the released energy density; the dark-energy density gives about 2 meV, far below the energy density needed for 1 MeV. (b) Temperature of the open interior after reheating ($\Gamma_\varphi = m$): the radiation cools as $1/t$ and never approaches nucleosynthesis. (c) Fraction of $\Delta V$ carried by the interior against the tilt.

## 5.3 Scope of the bound: what is excluded by the model and what only by energy conservation

The bound of Eq. (13) is a statement about a locally homogeneous conversion of a vacuum-energy difference that does not exceed the dark-energy density. It is not a universal prohibition on high-temperature bubble interiors, and it is stated here with its assumptions: (A1) one real scalar field minimally coupled to gravity, with the potential of Section 2; (A2) the false-vacuum energy is the observed dark energy; (A3) the interior is homogeneous and isotropic on open slices; (A4) no inflationary stage inside the bubble; (A5) all energy densities are sub-Planckian. Models

that violate one of these need a separate analysis. Table 9 lists the possible additional energy sources and says which of them the model excludes by more than energy conservation.

The geometric argument is the following. In the open slicing the curvature parameter $Y \equiv a^2\rho/(3M_{\rm Pl}^2)$ obeys $(da/dt)^2 = 1 + Y$. A hot big bang at nucleosynthesis needs $Y \sim 1$: in standard radiation domination $Ht = 1/2$, that is $\rho t^2 = 3M_{\rm Pl}^2/4$. In the curvature-dominated phase $a = t$ and $Y = \rho t^2/(3M_{\rm Pl}^2)$ can only decrease, as $1/t$ for matter and $1/t^2$ for radiation, unless the interior inflates. The energy density of the condensate and its decay products obeys the envelope

$$\rho(t) \le \epsilon\,\Delta V\,(mt)^{-3} \quad (t > t_0 = 1/m), \qquad Y(t_0) = \frac{\epsilon\,\Delta V}{3M_{\rm Pl}^2 m^2} \tag{14}$$

because matter dilutes more slowly than radiation and decay only moves energy from the slower to the faster component (checked against the closed form to machine precision, check V17). The mass at the true minimum is $m = \kappa_m \Delta V^{1/4}$ with $\kappa_m = \lambda^{1/4}[(3x_t^2 - 1)/2]^{1/2}\delta^{-1/4}$, which lies between 0.50 and 4.1 over the whole prior, so $Y(t_0) = c\sqrt{\Delta V}/M_{\rm Pl}^2$ is tiny for any sub-Planckian energy: $Y(t_0) = 2.0 \times 10^{-63}$ at $\Delta V = V_f$, and $Y(t_0) = 1$ would need $\Delta V/M_{\rm Pl}^4 = 1.8 \times 10^5$ at the benchmark, which is super-Planckian. Therefore, for any $\Delta V$ below the Planck scale, the interior is a Milne universe at all times, $Y$ never approaches unity (Fig. 9), and the energy density at one second after nucleation cannot exceed $\epsilon\Delta V/(m \cdot 1\,\mathrm{s})^3$. Requiring $\rho(1\ \mathrm{s}) \ge \rho_{\rm BBN} = (\pi^2 g_*/30)(1\ \mathrm{MeV})^4$ ($g_* = 10.75$) demands $\Delta V^{1/4} \ge 5.1 \times 10^{61}$ GeV, that is 42, 43 and 45 orders of magnitude above the Planck mass (for the smallest $\kappa_m$ in the prior, the benchmark and the largest). For the dark energy the envelope gives $\rho(1\ \mathrm{s}) = 4.4 \times 10^{-74}\,\rho_{\rm BBN}$.

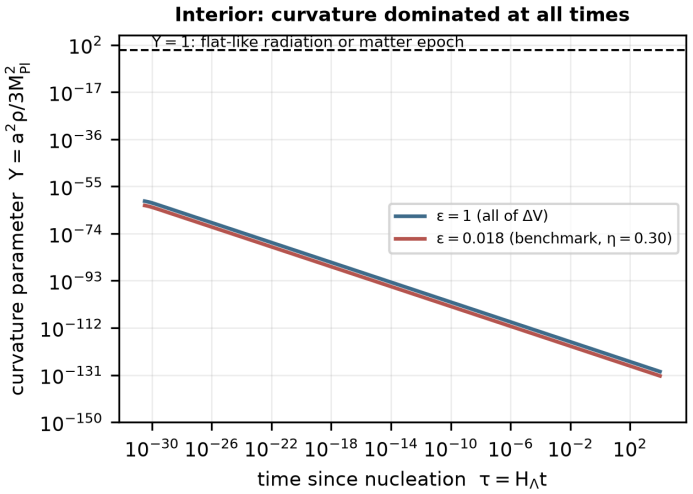



Figure 9: Curvature parameter $Y = a^2\rho/3M_{\rm Pl}^2$ of the bubble interior against the time since nucleation for $\Gamma_\varphi = m$. A flat-like radiation- or matter-dominated epoch needs $Y \sim 1$; the interior starts at $Y \sim 10^{-63}$ and moves away from it.

Three points follow. First, the anti-de Sitter case shows that energy conservation alone is not sufficient: a very deep true vacuum releases enough energy for 1 MeV, but its open interior then collapses within seconds, and the longest-lived interior has $T \lesssim 4$ meV. Second, the geometric statement is independent of the energy scale and of the value of $V_f$, and therefore covers an

Table 9: Energy sources and assumptions behind the reheating bound, and what excludes each.

| Source or assumption | What bounds it | Quantitative statement |
|---|---|---|
| Released vacuum energy, $V_t \geq 0$ | energy conservation, $\Delta V \leq V_f$ | $T_{\rm reh} \leq 2.19$ meV |
| Wall energy $(1-\epsilon_{\rm in})\Delta V$ | already counted in $\Delta V$; collisions redistribute it | benchmark: 98% of $\Delta V$ in the wall; the total is unchanged |
| Deeper true vacuum, $V_t < 0$ (anti-de Sitter) | geometry: crunch at $t_c = \pi\sqrt{3}M_{\rm Pl}/\sqrt{\lvert V_t\rvert}$ and the curvature envelope | $\lvert V_t\rvert = 1.4\times10^{35}\,V_f$ and a crunch after 4.6 s; a life of 13.8 Gyr needs $\lvert V_t\rvert \leq 15.7\,V_f$ and $T \leq 4.4$ meV |
| Interior inflation | the potential has no plateau above $V_f$; field range $\Delta\varphi/M_{\rm Pl} \sim 10^{-30}$ | at most $9.3\times10^{-62}$ e-folds (largest over four tilts) |
| Larger parent scale ($V_f \gg \rho_\Lambda$) | geometry: the envelope $\rho \leq \epsilon\Delta V(mt)^{-3}$ | $\rho(1\text{ s}) \geq \rho_{\rm BBN}$ needs $\Delta V^{1/4}$ 42 to 45 orders above $M_{\rm Pl}$ |
| Extra reservoir (other fields, radiation) | not part of the model; subject to the same curvature constraint | $Y \sim 1$ at $t = 1/m$ needs $\rho_x = 9.1\times10^{60}\,\Delta V$ |
| Inhomogeneity, bubble collisions | not included (A3); total energy is bounded by $\Delta V$ | local concentration possible; the average is unchanged |

additional reservoir too: it cannot raise $Y$ above the curvature bound unless its energy density exceeds $\Delta V$ by $9.1\times10^{60}$. Third, the escape routes are exactly those the model lacks: a long inflationary stage inside the bubble, a parent energy scale close to the Planck scale, or an additional mechanism that raises $Y$ at $t_0$. A successful cosmology would have to lie in that outside class.

A second scalar field $\chi$ does not change this. For any number of scalar fields with a potential $V(\varphi,\chi)$, the energy released in the transition of a homogeneous region is the potential difference between the false and true vacua, $V_f - V_t$, and the false-vacuum energy $V_f = V(\varphi_f,\chi_f)$ is the whole observed dark energy $\rho_\Lambda$: any energy stored in $\chi$ while it sits in the false vacuum gravitates as dark energy and is already counted in $V_f$. A larger release needs $V_t < 0$ (the anti-de Sitter row of Table 9, a crunch within seconds), and the curvature argument above is independent of the field content. A reservoir that is not vacuum energy (radiation, matter, kinetic energy of $\chi$) would have been present and gravitating throughout the cold era, where observations limit it to a fraction of the critical density, whereas the interior needs $\rho_x = 9.1\times10^{60}\,\Delta V$. A coupled two-field model is therefore a relabelling of $\Delta V$ and not a source of new energy, unless the second sector inflates or carries a Planck-scale parent energy, which is the outside class of Section 7.

### 5.4 Other fates and the singularity question

If the true vacuum has negative energy ($V_t < 0$) the open interior collapses to a singularity in a finite time, $\tau_c = \pi/\sqrt{\lvert V_t\rvert/V_f}$: $\lvert V_t\rvert = 0.01\,V_f$: 547 Gyr; $\lvert V_t\rvert = 0.1\,V_f$: 173 Gyr; $\lvert V_t\rvert = 1\,V_f$: 55 Gyr; $\lvert V_t\rvert = 10\,V_f$: 17 Gyr. A decay to a de Sitter vacuum with $V_t > 0$ leaves the interior with an accelerating phase and no radiation epoch. Only $V_t \simeq 0$ leaves a regular, expanding, but empty interior. No case yields a nonsingular bounce from the parent geometry: the bubble is a region of the parent de Sitter space, causally connected to it, and the Milne interior is geodesically complete only because it has no matter.

The distinction between daughter regions and a cyclic cosmology (Section 1) applies to these fates. The nucleated region is a daughter region in the sense of Option A: a cold, curvature-dominated open region bounded by the wall inside the parent de Sitter space and still causally connected to it, not a new hot universe. The potential has no further metastable minimum at the true vacuum, so repeated transitions of the interior would need a different potential; the entropy produced in reheating is bounded by the energy $\Delta V$ and is carried by an open interior of infinite comoving volume, so no entropy statement can be made for the daughter as a closed system. A cyclic cosmology (Option B) needs a nonsingular bounce, which in spatially flat general relativity with conventional matter requires a violation of the null energy condition ($\dot{H} = -(\rho + p)/2M_{\rm Pl}^2 > 0$) and raises problems of stability and entropy [9, 10]; the present model contains no such mechanism. These statements are qualitative: entropy accounting and the stability of an interior bounce are not computed.

## 6 Matter dilution and decoherence as a trigger

The original motivation for a late-time transition was that dilution of matter reduces environmental decoherence and thereby permits a macroscopic quantum transition. In the baseline model nothing of the kind appears: the potential $V(\varphi)$, hence $B$, does not depend on the matter density, and the Gibbons–Hawking temperature $T_H = H/2\pi = 1.9 \times 10^{-43}$ GeV [43] is $6 \times 10^{-32}$ of the field scale; thermal-type corrections to $B$ are of relative size $\sim g\,(\kappa \simeq 3.5)$, that is $10^{-59}$ (Section 3.4). A dependence on matter must be introduced by a coupling, and a conclusion that held only for one coupling would be weak. The requirement can be stated without choosing one.

Any coupling of non-relativistic matter of mass $m_{\rm mat}(\varphi)$ adds $\rho_m \ln m_{\rm mat}(\varphi)$ to the effective potential, so that the energy difference between the vacua changes by $\rho_m\,\Delta \ln m$, where $\Delta \ln m$ is the fractional change of the matter mass between the two field values and $\rho_m = (\Omega_m/\Omega_\Lambda)V_f$ today. Write $\varepsilon_m \equiv \rho_m \Delta \ln m / \Delta V$ for the relative shift. To first order in the tilt shift $\Delta\eta$,

$$\frac{\Delta\eta}{\eta} = \frac{2\,\delta(\eta)}{\eta\,(x_f - x_t)}\,\varepsilon_m, \qquad \Delta B = B \left|\frac{d\ln b}{d\ln \eta}\right| \frac{\Delta\eta}{\eta} \tag{15}$$

with $\delta(\eta) = \Delta V/(\lambda f^4)$ of Eq. (3). A change of the rate by a factor $e$ needs $\Delta B = 1$; moving a benchmark from $B = 441$ to the critical action $B_c = 287$ needs $\Delta B = 154$ ($\Delta B = 13.7$ for $\eta = 0.32$). With $|d\ln b/d\ln\eta| = 5.5$ at $\eta = 0.30$ this gives $\Delta \ln m = 8.9 \times 10^{-4}$ and $1.4 \times 10^{-1}$ respectively: the matter mass must differ by between about 0.1% and 10% between the two vacua. Table 10 translates this to specific couplings.

The conclusion is therefore conditional. For the simplest linear coupling the required strength lies 26 to 28 orders of magnitude above what unscreened laboratory and solar-system tests allow. Quadratic or screened couplings are not excluded by this paper: they evade local tests by construction, but they are tuned, because they require a coupling scale of the same size as the dark-energy potential and a mass change of at least 0.1% of the matter between vacua. The dilution of matter can act as a trigger only through such an added, tuned structure, not through the framework itself. A noise-assisted mechanism would need an explicit open-system influence

Table 10: Requirement on a matter coupling that changes the rate by a factor $e$ (left end) or moves the benchmark into the reset window (right end).

| Coupling | Requirement | Comment |
|---|---|---|
| General | $\Delta \ln m \geq 8.9 \times 10^{-4}$ to $1.4 \times 10^{-1}$ | independent of the coupling function; $\eta = 0.30$ |
| Linear, $m = m_0 e^{\beta\varphi/M_{\rm Pl}}$ | $\beta \geq 3.7 \times 10^{26}$ to $5.7 \times 10^{28}$ (for $\eta = 0.32$: $4.7 \times 10^{26}$ to $6.4 \times 10^{27}$) | $\Delta\varphi/M_{\rm Pl} = 2.4 \times 10^{-30}$; unscreened tests restrict $\beta \lesssim O(1)$; screening (chameleon type) lowers the local coupling, not the cosmological one [35] |
| Quadratic, $m = m_0(1 + \varphi^2/2\Lambda_q^2)$ | $\Lambda_q \leq 58$ to 5 meV ($\eta = 0.30$) | $f = 3.0$ meV, so $\Lambda_q \sim$ (2–20)$f$: a new interaction at the scale of the potential; symmetron-type screening [36] can hide it in dense environments |
| Density-dependent effective potential | same $\Delta \ln m$ | the model is then specified by $\rho_m F(\varphi)$ with $F$ changing by that amount across $\Delta\varphi$ |

functional and a calculated change of the tunneling exponent, which is not assumed here.

# 7 Consistency conditions and falsification tests

A model of this class must satisfy internal consistency conditions, which are mathematical or follow from the framework, and it can be confronted with observational tests. The two are kept apart because failing the first means the model is ill-defined, while failing the second means that it is disfavored by data. All are computed in this paper and can be recomputed for a different potential with the supplied code.

## 7.1 Internal consistency conditions

- C1 (existence). A metastable minimum requires $\eta < \eta_c = 0.385$; for larger tilt there is nothing to decay.
- C2 (semiclassical validity). The loop expansion parameter $1/B$ must be small. All draws with $B < 30$ (2.5% of the prior) have rates far above the survival bound, so the classification of allowed draws is unaffected (Section 4.3).
- C3 (negative mode and flat limit). The bounce has one negative mode and gravity enters as an $O(g)$ perturbation, $g \sim 10^{-60}$ (Section 3.3).
- C4 (energy). With $V_t \geq 0$ the released energy density cannot exceed $\rho_\Lambda$, so $T_{\rm reh} \leq 2.2$ meV; with $V_t < 0$ the interior crunches (Section 5.3).
- C5 (geometry). For any sub-Planckian energy the open interior stays curvature dominated, $Y(t) \leq Y(t_0) = 2.0 \times 10^{-63}$, unless it inflates; the single-field potential gives at most $9.3 \times 10^{-62}$ e-folds (Section 5.3).

- C6 (percolation). A global transition requires $\Gamma/H^4 = O(1)$ or larger, a window of width $\Delta B = 3.9$ below the survival bound that holds 0.12% of the prior volume.

- C7 (thermal consistency). A scalar in equilibrium with the CMB at redshift $z$ keeps its false minimum only if $c < \theta_c \lambda f^2 / T_\gamma(z)^2$, i.e. $c < 1.1 \times 10^{-5}$ for a bath coupled back to recombination, and the survival condition then applies to the rate history $\Gamma(T_\gamma(t))$ (Section 4.8).

### 7.2 Observational tests

- O1 (survival). The absence of a bubble in our past light cone requires $\Gamma V_4 < 3.0$, i.e. $\Gamma/H^4 < 51$ at 95% confidence, which excludes 15% of the prior volume of the benchmark family and constrains the future mean decay time only to $t_{\rm dec} \gtrsim 8.2 \times 10^7$ yr.

- O2 (reset window). A transition on cosmological time scales requires $B_c \lesssim B \lesssim B_c + 28$ for decay within $10^{20}$ yr; this holds for 0.77% of the prior volume. An action above $B_c + 212$ puts the decay beyond $10^{100}$ yr.

- O3 (hot epoch). A hot universe in the nucleated region requires $\rho_{\rm reh} \gtrsim (1\ {\rm MeV})^4$, which is $1.4 \times 10^{35}$ times the dark-energy density; the bound of Section 5 excludes it for decay of the present dark energy in this framework.

- O4 (trigger). Matter dilution as the trigger needs $\Delta \ln m \gtrsim 10^{-3}$ between the vacua, a linear coupling of $\beta \sim 10^{26}$–$10^{29}$ or a screened coupling at a scale of tens of meV (Section 6), which local gravity and fifth-force tests can probe if the screening is incomplete.

- O5 (signals). A solitary transition produces no signal in our patch; bubble collisions [25, 33, 34] require a population and $\Gamma/H^4 = O(1)$ (C6). The vacuum is indistinguishable from a cosmological constant ($w = -1$) until it decays. A measured evolution of the dark energy, which recent baryon-acoustic-oscillation analyses combined with the CMB and supernovae prefer at the level of a few standard deviations depending on the supernova sample [44], would favor a rolling field and not a metastable vacuum. No observational discriminant of the cyclic reset is available from this model.

- O6 (cooling). If the scalar is coupled to the CMB the decay probability was larger in the past and falls as the universe cools: equilibrated scenarios with $c \geq 0$ lift the excluded prior mass from 15% to 24–49% and leave no survival-consistent draw that decays within $10^{20}$ yr, so the absence of a transition so far constrains the coupling of the scalar to the CMB as well as its potential (Section 4.8).

A transition producing a successful cosmology would have to lie outside the class studied here. It would need a parent vacuum energy of at least $\sim 10^{34} \rho_\Lambda$ (an earlier, higher-energy phase of the universe rather than the present one) and, by Section 5.3, a post-nucleation region that inflates before it reheats, so that the curvature term and the horizon are handled, and a mechanism to restore the high-energy parent from the late universe. The cyclic models that have been

constructed provide such mechanisms by other means [4, 5]; this paper shows that tunneling of the present dark energy does not.

## 8 Discussion and limitations

- A single scalar field and a quartic-tilt potential are used. Other potentials change $b(\eta)$ and the interior energy fraction, but not the scales: the rate depends on the combination $M_{\rm Pl}^4/V_f$ and the interior energy on $V_f$, both fixed by the data, so O1, O3 and the geometric argument of Section 5.3 are generic.

- The one-loop prefactor is a systematic uncertainty, not a result; the conclusions are insensitive to it over six decades (Section 4.3).

- The decay is treated as a flat-space process because $g \sim 10^{-60}$; the gravitational code is validated against analytic limits and used for the map of the regime in which gravity matters, which the dark-energy scale excludes.

- The survival criterion assumes a constant rate and a false vacuum throughout the history of the observed universe; a field that evolved or was thermally affected in the early universe would change the early part of the cone, which carries a small part of $V_4$.

- The interior evolution assumes a homogeneous open slicing and a perturbative decay of the condensate to radiation; bubble-wall collisions can concentrate energy locally, but the average energy density is bounded by $\Delta V$ (Table 9).

- Up-tunneling from the false to a higher vacuum, the decay of the true vacuum in curved space [45], thermal activation of the Hawking–Moss type [30] and decays of the present vacuum into other vacua are not considered.

- The priors of the Monte Carlo are documented assumptions, not measurements; the regime fractions describe the sample space of the model, not the probability of a transition in our universe.

- The matter-coupling analysis is first order in the tilt shift and does not simulate any specific screened model.

- The thermal analysis assumes equilibrium with the CMB (an upper bound on the thermal channel), the leading high-temperature term with one coefficient $c$, and a bath coupled below a stated redshift; it neglects higher-order thermal corrections and the interpolation between the O(4) and O(3) saddles near $T_\times$, which a periodic-instanton calculation would refine.

- No two-field simulation of the post-nucleation evolution is performed: the argument of Section 5.3 shows that a second field cannot supply energy, and the repeated-transition and entropy questions of Section 5.4 are stated but not computed.

# 9 Conclusions

A late-time false-vacuum transition of the present dark energy is a well-defined calculation, and for the explicit potential studied it can be done to a stated precision. The quantum decay is a flat-space process controlled by the action $B = b(\eta)/\lambda$, which must lie near $B_c \approx 287$ for the transition to be neither excluded nor effectively absent; with survival evaluated on the past light cone the bound is weak ($\Gamma/H^4 < 51$) and the window is set by the action itself. Cosmological cooling does not help: a thermal mass rescales the potential exactly, the thermal O(3) channel dominates above 0.18 meV and falls monotonically as the universe cools, and a scalar in equilibrium with the CMB is more strongly constrained (24–49% of the prior excluded against 15%) with no late-time decay window left. The nucleated daughter region carries at most the dark-energy density, reheats to at most 2.2 meV and cools as $1/t$ in a curvature-dominated open universe; this rests on geometry and not on energy conservation alone, holds for any sub-Planckian parent energy and any number of scalar fields, and a trigger by matter dilution would need a change of the matter mass between the vacua of at least 0.1%. A daughter region of the present dark energy is cold and empty, and the cyclic-reset hypothesis, in the form tested, is not supported; the framework identifies what would have to change.

# Declarations

**Funding.** This research received no specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

**Conflict of interest.** The author declares no competing interests.

**Ethics approval, consent to participate, consent for publication.** Not applicable.

**Data availability.** The data underlying every figure and table (bounce tables, error budget, survival calculation, falsification map, Monte Carlo summary, interior evolution, scope and coupling estimates, verification results) are deposited with the code at https://github.com/sandlerleon/late-time-vacuum-decay and archived at https://doi.org/10.5281/zenodo.23752471.

**Code availability.** All code (Python, NumPy, SciPy, Matplotlib) is released under the MIT licence at https://github.com/sandlerleon/late-time-vacuum-decay, archived at https://doi.org/10.5281/zenodo.2
Running the scripts in the stated order regenerates every number in this paper, and verify.py runs 21 executable checks. The manuscript is archived at https://doi.org/10.5281/zenodo.23752470.

**Authors' contributions.** L.S.: conceptualization, methodology, software, formal analysis, investigation, visualization, writing.

**Use of generative AI.** During the preparation of this work the author used Claude (Anthropic) to assist with the design of the numerical analysis, writing of the code, literature checking and drafting. The author reviewed and edited the content, checked the code against analytic limits and executable tests, verified the cited references, and takes full responsibility for the publication. No AI tool is an author.

## References